Violet(Zimeng) Ye

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Education

University of California, Berkeley Expected May 2028

B.A. Computer Science & B.A. Economics (Intended), GPA: 3.88 / 4.0

Relevant Coursework: Data Structures & Algorithms, Linear Algebra & Differential Equations, Principles & Techniques of Data Science, Structure & Interpretation of Computer Programs, Probability & Statistics, Multivariable Calculus, Econometrics-Math Intense, Full-Stack Web Development, Intro to Quant Finance

Experience

iPick.AI – Machine Learning Engineering Intern Aug 2026 – Present

- Building an LLM-powered **portfolio optimization tool** (Claude/OpenAI APIs) that analyzes user holdings and generates risk-weighted investment recommendations
- Implementing a **reinforcement learning system** that uses weekly stock-performance feedback as reward signals to iteratively improve ranking of high-growth and high-risk equities across **5000+ ticker universe**

UC Berkeley – Undergraduate Research Assistant (Data Engineering) Apr 2026 – Present

- Developed a multi-source **ETL pipeline** (Apps Script + Python) to collect UPC-level nutrition data for **50K+ food products** from USDA, OFF, and embedding-based name matching, verified by rule-based consistency checks
- Integrating the **OpenAI API** to locate and read nutrition labels from product images, and training a **match-scoring model** on reviewer decisions of uncertain matches, raising precision and reducing review time

Atypica.AI, Tezign – Software Engineering Intern & AI Research Intern Mar 2026 – Sep 2026

- Shipped **Atpio**, a full-stack **Next.js/TypeScript platform** that turns a natural-language brief into an embeddable feedback widget using LLM-generated form schemas and REST APIs, with a PII-redaction layer so response data stays anonymous
- Built a leakage-controlled **evaluation pipeline** in Python to test whether AI personas reflect real consumers; across **1,153 respondents**, persona scores correlated only weakly with held-out behavior, and **bge-m3 + K-means** analysis of **182 personas** benchmarked their diversity against human and generic baselines
- To diagnose these failures, reproduced and extended **Anthropic's J-space interpretability method** on an open-source 8B model, building a layer-wise probing harness (paper in preparation for **CHI**)

China–U.S. Green Fund Management – Data Science Intern Jul 2024 – Aug 2024

- Automated data collection and cleaning with **Python** for performance data across commercial solar and green-energy projects, replacing manual spreadsheet work and speeding up reporting
- Merged data sources into a standardized dataset and built **time-series and regression forecasts** of green-energy investment trends, which fed my team's ESG industry reports

Projects

End-to-End News Recommendation System | *scikit-learn, FastAPI, MLflow* Sep 2026 – Present

- Built a data pipeline over Microsoft's MIND impression logs, turning exposures into labeled click samples
- Implementing an end-to-end baseline from popularity retrieval to logistic regression ranking with **NDCG/MRR evaluation** and **MLflow** tracking; serving via a **FastAPI** endpoint with request logging and **PSI drift checks**
- Extending retrieval with **item-based CF** and **FAISS** embedding search, and upgrading the ranker toward LightGBM and Wide & Deep

PackSmart | *Node.js, JavaScript, REST APIs, OpenAI API, Hugging Face* Mar 2026 – Present

- Built a **full-stack travel outfit planner** (single-page app + **Node.js REST backend**) that recommends daily outfits from a user's uploaded closet, trip activities, and weather, using rule-based scoring plus OpenAI API calls
- To replace API calls with smaller model, developing a **distillation data pipeline** that generates synthetic closet/trip scenarios, labeled by a large **teacher model** under a strict JSON schema, and filters outputs with rule-based validators

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, R
Tools: Git, VS Code, Figma, IntelliJ, Cursor, Claude Code, Codex, Jupyter Notebook, Vercel, Supabase, Hugging Face
Skills: Machine Learning, EDA, Probability, Regression, React, Next.js, Node.js, Express.js, REST APIs

Awards & Activities

Awards: AIME 11/15, qualified for **USAMO**; HiMCM Finalist, Published **Scopus Q1 Journal**; NEC National 1st
Leadership: Team Lead, GreenDesert Public Welfare Group; Production Chair & Producer, T.I.n.Y Chinese Theater Group; Founder, School Bulletin Board Platform